Eva Martín López

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Experience

UC Irvine, Balsells Research Fellow – Visiting Researcher
Irvine, CA
Jan 2026 – Present

- Conducting Master's thesis research in Professor Pierre Baldi's group
- Applying deep learning to biomedical imaging, focusing on chemoresistance prediction at the single-cell level in breast cancer research
- Developing and training convolutional neural networks for medical image analysis
- Awarded the Balsells Fellowship, a competitive scholarship for Catalan researchers at UCI

Geprom (Telefónica Tech), Software Developer
Castelldefels, ES
Jan 2025 – Feb 2026

- Developed data spaces for secure data exchange in industrial environments using big data technologies and cloud-based systems
- Contributed to industrial automation and digital transformation projects (Industry 4.0)
- Focused on data infrastructure and industrial digitalization

Institute for Bioengineering of Catalonia (IBEC), Research Assistant in Data Science
Barcelona, ES
Jun 2023 – Jan 2025

- Collaborated on research projects related to data analysis, visualization, and machine learning for biomedical research
- Focused on data preprocessing, statistical modeling, and the application of AI in healthcare
- Co-authored a publication in *Critical Care Medicine* on breathing effort and survival in ARDS patients

Janssen (Johnson & Johnson), Internship – Computational Chemistry Dept.
Toledo, ES
Jan 2023 – Jul 2023

- Developed a Python package for the automation of molecular docking simulations
- Worked with Git and Linux in a pharmaceutical R&D environment

Education

MS **Universitat Politècnica de Catalunya (UPC)**, Master's in Data Science
Barcelona, ES
Sep 2024 – Jun 2026

- Thesis at UC Irvine: Deep learning for biomedical imaging (Balsells Fellowship)
- Focus: big data, statistical modeling, process mining, and machine learning

BS **ESCI – Universitat Pompeu Fabra**, Bachelor's in Bioinformatics
Barcelona, ES
Sep 2020 – Jun 2023

- Santander ESCI-UPF Excellence Scholarship (2020–2021 and 2021–2022)

Publications

Association of Breathing Effort With Survival in Patients With Acute Respiratory Distress Syndrome Jul 2025

Critical Care Medicine

DOI: 10.1097/CCM.0000000000006477

Predictive Model for High Flow Nasal Cannula Failure Based on Metabolomics and Clinical Data Under review

Journal of Critical Care (submitted 2025)

Conferences

Conference of the Catalan Society of Intensive and Critical Medicine (SOCMIC)

- Barcelona, Spain (February 2024). Contributed to the publication and received the **Best Communication Award**

Annual Conference of the Spanish Society of Biomedical Engineering (CASEIB)

- Cartagena, Spain (November 2023). Poster presentation

ISOCS Winter School – Chemical Sensing for Biomedical Applications

- Bormio, Italy (2024). Winter school on transitioning chemical sensing from proof of concept to medical devices

Awards and Scholarships

Balsells Fellowship, UC Irvine (2026)

- Competitive scholarship for Catalan researchers to conduct research at the University of California, Irvine

Best Communication Award, SOCMIC Conference (2024)

- Awarded for the publication contribution at the Conference of the Catalan Society of Intensive and Critical Medicine

Santander ESCI-UPF Excellence Scholarship (2020–2021, 2021–2022)

- Merit-based scholarship for academic excellence

Additional Training

- **Summer School at UPC (2023):** Applied Machine Learning to Solve Real-Life Problems; Web Application Development with Shiny
- **Summer School at Eigenvector Research (2023):** MATLAB Course in Machine Learning
- **Classical Ballet Conservatory:** Institut del Teatre, Barcelona (2014–2017)

Languages

- **Spanish and Catalan:** Native
- **English:** Advanced

Technologies

- **Programming:** Python, R, SQL, MATLAB, Bash
- **Libraries & Frameworks:** PyTorch, Scikit-learn, NumPy, Pandas, Shiny
- **Tools & Platforms:** Git, Linux, Docker, Cloud Technologies, LaTeX
- **Areas:** Machine Learning, Deep Learning (CNNs), Statistical Modeling, Data Preprocessing, Biomedical Imaging